

Introduction to Test & Measurement in Physical Education

This cheatsheet covers fundamental fitness tests, methods for computing Basal Metabolic Rate (BMR), and the Rikli & Jones Senior Citizen Fitness Test. Understanding these measurements is crucial for assessing physical fitness across different age groups and populations.

Fitness Test – SAI Khelo India Fitness Test in School

Age group 5-8 yrs/ class 1-3

For primary school children, the focus is on acquiring Fundamental Movement Skills (FMS), which are building blocks for physical activities.

- Body Composition: Measured by Body Mass Index (BMI).
- Coordination: Assessed by the Plate Tapping Test.
- Balance: Evaluated using the Flamingo Balance Test.

Age group 9-18 yrs/ class 4-12

For older students, overall physical fitness components are assessed.

- Body Composition: Measured by Body Mass Index (BMI).
- Speed: Assessed by the 50mt Speed Test.
- Cardiovascular Endurance: Evaluated with the 600mt Run/Walk.
- Flexibility: Measured by the Sit & Reach Flexibility Test.
- Strength Test:
 - Abdominal Strength: Partial Curl-Up.
 - Upper Body Strength: Push-Ups for boys, Modified Push-Ups for girls.

Body Mass Index (BMI)

- Definition: Body Composition refers to the distribution of muscle and fat. BMI is a measure based on weight and height.
- Formula: $BMI = \frac{Weight(kg)}{Height^2(m^2)}$
 - W: Body weight in kilograms.
 - H: Height in meters.
- Interpretation: Higher scores usually indicate higher levels of body fat.
- Equipment Required: Flat, clean surface, weighing machine, stadiometer/measuring tape.

BMI Categories

- Underweight: < 18.5
- Normal: 18.5 - 24.9
- Overweight: 25 - 29.9
- Obese: 30 - 34.9
- Extremely Obese: > 35



Plate Tapping Test

- Purpose: Measures speed and coordination of limb movement.
- Equipment: Adjustable height table, 2 yellow discs (20cm diameter), rectangle (30x20 cm), stopwatch.
- Procedure:
 1. Adjust table height for comfortable standing.
 2. Place two yellow discs 60cm apart on the table.
 3. Place the rectangle equidistant between discs.
 4. Non-preferred hand rests on the rectangle.
 5. Preferred hand taps back and forth between discs over the middle hand as quickly as possible.
 6. Repeat for 25 full cycles (50 taps).
- Suggestions: Encourage balanced posture, feet shoulder-width apart. Maintain constant pace.

Flamingo Balance Test

- Purpose: Measures the ability to balance successfully on a single leg, assessing leg, pelvic, and trunk muscle strength, and static balance.
- Equipment: Non-slippery even surface, stopwatch, beam (or just standing on the floor).
- Procedure:
 1. Stand on the beam. Instructor may assist with balance initially.
 2. Flex free leg at the knee, holding foot close to buttocks.

3. Start stopwatch when instructor releases participant.
4. Pause stopwatch each time balance is lost (falling off beam or releasing foot).
5. Resume timing until balance is lost again.
6. Count falls in 60 seconds. Test terminates if >15 falls in the first 30 seconds.

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50m Standing Start

- Purpose: To determine speed.
- Equipment: Measuring tape or marked track, stopwatch.
- Procedure:
 1. Thorough warm-up including practice starts.
 2. Start from a stationary standing position (hands cannot touch the ground), with one foot in front of the other.
 3. Run a single maximum sprint over 50 meters.
 4. Time is recorded.

600 Metre Run/Walk

- Purpose: To measure endurance.
- Equipment: 600 metre track, stopwatch.
- Procedure:
 1. Subject stands behind the starting line.
 2. At the signal 'GO', subject starts running/walking 600 yards.
 3. Time is recorded in minutes and seconds.

Sit and Reach

- Purpose: To measure the flexibility of the hip region, including the back and hamstrings.
- Equipment: Sit and reach box.
- Procedure:
 1. Sit on the floor with legs stretched straight, shoes removed.
 2. Soles of feet flat against the box, knees locked and pressed flat.
 3. Palms facing downwards, hands on top of each other or side by side.
 4. Reach forward along the measuring line as far as possible, keeping hands level.
 5. Hold the reach for 1-2 seconds; distance is recorded. No jerky movements.

Partial Curl-Up

- Purpose: To test the strength and endurance of abdominal muscles.
- Equipment: Flat, clean, cushioned surface, recording sheet.
- Procedure:
 1. Lie on back, knees flexed, feet 12 inches from buttocks (feet not held).
 2. Arms extended, resting on thighs. Head in neutral position.
 3. Curl up slowly until shoulders come off mat two inches, then back down.
 4. One complete curl-up takes 3 seconds (1.5 up, 1.5 down, no hesitation).
 5. Continue until exhaustion, maintaining rhythm and continuous abdominal engagement.

Push-up Test

- Purpose: To measure upper body strength and endurance.
- Equipment: Floor mat, record sheet.
- Procedure:
 1. Start with hands and toes on floor, body straight, feet slightly apart, arms shoulder-width apart and extended.
 2. Lower body until elbows are at a 90-degree angle or body touches a predetermined point.
 3. Return to starting position.
 4. Repeat without rest until exhaustion or target number is reached, maintaining rhythm.

Modified Push-Up (for girls)

- Purpose: To measure upper body strength and endurance.
- Equipment: Set and reach box (this seems incorrect based on typical modified push-up description, usually just a mat is needed).
- Procedure:

1. Begin in a kneeling position on a mat, hands below shoulders, knees behind hips. Back angled and long.
2. Tuck toes under, tighten abdominals.
3. Bend elbows to lower chest toward floor, keeping gaze in front of fingertips.
4. Press chest back up to start position.
5. Repeat for desired number of repetitions.

Computing Basal Metabolic Rate (BMR)

- Definition: BMR is the number of calories the body needs to perform basic life-sustaining functions (breathing, circulation, nutrient processing, cell production). 60-75% of daily calories are burned this way.
- Significance: It's the metabolism rate when waking up after fasting during sleep. BMR provides energy for vital organs. Lean muscle mass increases BMR, but with limits.
- Harris-Benedict Formula:
 - For Women: $BMR = 655 + (9.6 \times \text{weight in kg}) + (1.8 \times \text{height in cm}) - (4.7 \times \text{age in years})$
 - For Men: $BMR = 66 + (13.7 \times \text{weight in kg}) + (5 \times \text{height in cm}) - (6.8 \times \text{age in years})$

Rikli & Jones - Senior Citizen Fitness Test

This test battery assesses functional fitness in older adults.

Chair Stand Test for Lower Body Strength

- Purpose: Measures lower body strength.
- Equipment: Chair with straight back and seat (at least 44 cm high), stopwatch.
- Procedure:
 1. Place chair against a wall.
 2. Sit in middle of seat, feet shoulder-width apart, flat on floor.
 3. Arms crossed at wrists, held close to chest.
 4. Stand completely up, then sit completely back down.
 5. Repeat for 30 seconds. Count total complete stands.
- Scoring: Number of complete stands in 30 seconds. Norms vary by age and gender.

Arm Curl Test for Upper Body Strength

- Purpose: Measures upper body strength and endurance.
- Equipment: 5 lb weight for women, 8 lb for men; armless chair, stopwatch.
- Procedure:
 1. Sit on chair.
 2. Curl the weight by fully bending the arm, then fully straightening it.
 3. The administrator's hand may be placed on the biceps, and the lower arm must touch the tester's hand for a full bicep curl.
 4. Repeat as many times as possible within 30 seconds.
- Scoring: Total number of curls in 30 seconds. Norms vary by age and gender.

Chair Sit & Reach Test for Lower Body Flexibility

- Purpose: Measures lower body flexibility.
- Equipment: Ruler, chair with straight back (approx. 44 cm height).
- Procedure:
 1. Sit on edge of chair (against a wall).
 2. One foot flat on floor, other leg extended forward with straight knee, heel on floor, ankle at 90°.
 3. Place one hand on top of the other, middle fingers flush.
 4. Inhale, then exhale and slowly reach forward toward toes, bending at hip, keeping back straight and head up. Avoid bouncing.
 5. Hold reach for 2 seconds.
 6. Measure distance between fingertips and toes. If fingertips touch toes, score is zero. Negative score if fingers don't reach, positive if they overlap.
 7. Repeat 2 times on each leg.
- Scoring: Recorded to nearest 1/2 inch or 1 cm. Norms vary by age and gender.

Back Scratch Test for Upper Body Flexibility

- Purpose: To measure the flexibility of the shoulder/upper body.
- Equipment: Ruler.

- Procedure:
 1. Stand.
 2. Place one hand behind head and back over shoulder, reaching as far down middle of back as possible, palm touching body, fingers downwards.
 3. With other hand, reach up behind back to try and touch/overlap fingers.
 4. Measure distance between fingertips.
- Scoring: Record the best score out of two trials to the nearest centimetre or 1/2 inch. Norms vary by age and gender.

Eight Foot Up & Go Test for Agility

- Purpose: Measures speed, agility, and dynamic balance.
- Equipment: Stopwatch, straight-back/folding chair (44 cm high), cone marker, measuring tape, clear area.
- Procedure:
 1. Place chair against a wall. Place marker 8 feet in front of chair.
 2. Subject sits fully, hands on knees, feet flat.
 3. On "Go," timing starts. Subject stands and walks (no running) as quickly and safely as possible around the cone and back to sit in chair.
 4. Timing stops when seated.
 5. Perform two trials.
- Scoring: Time taken in seconds. Norms vary by age and gender.

Six Minute Walk Test for Aerobic Endurance

- Purpose: Assesses aerobic fitness or endurance for activities like walking, stair climbing.
- Equipment: Measuring tape, stopwatch.
- Procedure:
 1. Mark a rectangular course (e.g., 45.72 m or 50 yards, or 20x5 yards with cones).
 2. Walk maximum distance as quickly as possible for six minutes.
 3. A practice trial is given. Participant can stop anytime.
- Scoring: Total distance covered in six minutes, recorded to the nearest metre/yard. Norms vary by age and gender.

